ORIGINAL ARTICLE

Integrated Digital Twin and Artificial Intelligence Architecture for Monitoring Water Distribution Systems

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Filled model article for demonstration of SETARI structure and layout.

Abstract

Water distribution systems require operational decisions based on hydraulic information, asset condition and measurement time series. This model article presents an integrated digital twin and artificial intelligence architecture for consolidating field data, representing system behavior and supporting the identification of operational deviations. The method is structured in five layers and includes acquisition, treatment, modeling, validation and analytics. In an illustrative dataset, the integrated configuration reduced normalized error by 54% compared with the baseline condition and maintained consistency between observed and estimated values. The results exemplify how quantitative evidence should be presented and discussed in manuscripts submitted to SETARI. It is concluded that the usefulness of a digital twin depends on data quality, model calibration and the incorporation of outputs into the decision-making process. The values reported here are for editorial illustration only and must be replaced by actual results in the submitted article.

Keywords: digital twin; artificial intelligence; automation; hydraulic systems; monitoring; applied engineering.

Resumo

Sistemas de distribuição de água demandam decisões operacionais baseadas em informações hidráulicas, estado de ativos e séries temporais de medição. Este artigo-modelo apresenta uma arquitetura integrada de gêmeo digital e inteligência artificial para consolidar dados de campo, representar o comportamento do sistema e apoiar a identificação de desvios operacionais. O método é estruturado em cinco camadas e inclui aquisição, tratamento, modelagem, validação e analítica. Em um conjunto de dados ilustrativo, a configuração integrada apresentou redução de 54% no erro normalizado em relação à condição de referência e manteve coerência entre valores observados e estimados. Os resultados exemplificam como evidências quantitativas devem ser apresentadas e discutidas em manuscritos submetidos à SETARI. Conclui-se que a utilidade de um gêmeo digital depende da qualidade dos dados, da calibração do modelo e da incorporação das saídas ao processo decisório. Os valores aqui apresentados têm finalidade exclusivamente editorial e devem ser substituídos por resultados reais no artigo submetido.

Palavras-chave: gêmeo digital; inteligência artificial; automação; sistemas hidráulicos; monitoramento; engenharia aplicada.

1. INTRODUCTION

Digital transformation has expanded the ability to monitor and operate engineering systems in near real time by integrating sensors, mathematical models and analytical tools. In this context, digital twins provide an integration framework between a physical asset and its computational representation, enabling operational state tracking, scenario testing and early detection of deviations. In water distribution systems, this approach is particularly relevant because operation involves hydraulic variables, availability constraints, losses, asset aging and decisions that must be made under uncertainty.

Despite the progress of supervisory control and data acquisition platforms, field information, technical registries, maintenance histories and hydraulic models are still frequently fragmented. Fragmentation reduces decision traceability and makes it difficult to convert operational data into actionable knowledge. The research problem of this model article is therefore how to structure an architecture capable of integrating sensor data, a hydraulic model and artificial intelligence algorithms in an auditable, reproducible and decision-oriented manner.

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The central contribution presented as an example is a five-layer architecture: instrumentation, data integration, physical model, intelligent analytics and decision support. The purpose is to demonstrate how a SETARI article should state the research gap, describe the method, report quantitative results, discuss limitations and record the conditions required for reproducibility. All numerical values used in this document are illustrative and must be replaced by actual data in the submitted manuscript. In industrial applications, practical relevance should be accompanied by measurable performance criteria. Operational, energy or reliability gains need to be reported with values, intervals and operating conditions.

The literature review should establish relationships among concepts, methods and limitations rather than present a sequence of disconnected summaries. In technology-oriented fields, it is important to distinguish proof-of-concept studies from those validated at operational scale. This distinction helps position contribution maturity and justify how the proposed solution adds knowledge beyond the implementation of existing tools. The methodological structure should also record software versions, parameters and preprocessing decisions. These elements are essential for reproducibility and for an adequate assessment of the technological contribution.

1.1. Research gap and contribution

Applied literature demonstrates the potential of integrating supervisory systems, physical models and machine-learning methods, yet many studies describe only part of the information cycle. The gap adopted in this example is the lack of an editorially traceable flow connecting acquisition, verification, model, indicator and decision. This formulation illustrates that an introduction should end with a verifiable contribution rather than merely presenting the topic. To ensure consistency, each variable should be associated with its unit, source, update frequency and validation rule. This description avoids ambiguity and makes the study auditable by external readers.

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The literature should also be analyzed according to the level of validation. Simulation-only studies may be appropriate for early exploration, but conclusions must match the level of evidence. Experimental or operational studies, in turn, should document test conditions and uncertainty in sufficient detail to enable comparison and reproduction. The methodological structure should also record software versions, parameters and preprocessing decisions. These elements are essential for reproducibility and for an adequate assessment of the technological contribution.

1.2. Paper organization

After this introduction, Section 2 presents the background; Section 3 describes materials and methods; Section 4 reports the results; Section 5 discusses the findings; Section 6 presents implementation implications; Section 7 states limitations; and Section 8 consolidates the conclusions.

2. BACKGROUND AND RELATED WORK

A digital twin can be understood as an updatable computational representation of a physical system, fed by data and used for analysis, prediction or decision support. In engineering, the mere existence of a model does not characterize a digital twin; authors must describe how updates occur, which data are incorporated and which decisions benefit from the representation. To ensure consistency, each variable should be associated with its unit, source, update frequency and validation rule. This description avoids ambiguity and makes the study auditable by external readers.

Artificial intelligence can complement the physical model in classification, anomaly detection, forecasting and estimation of unmeasured variables. However, high predictive performance does not replace physical validation. Data-driven models should be compared with baselines and tested on independent sets while preserving separation between training, validation and test data. The narrative should clearly separate observation, hypothesis and interpretation. When comparisons with previous work are made, the text must state the basis of comparison and avoid superiority claims without equivalent evidence.

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2.1. Comparative synthesis

Table 1 illustrates a synthesis of related work. In actual submissions, each row should represent a source that is effectively cited, and the final column should highlight the difference introduced by the proposed work.

Approach	Data	Validation	Main limitation
Physical model	SCADA	Measurements	Manual update
Machine learning	Historical	Hold-out	Low interpretability
Integrated architecture	Real time	Cross + physical	Higher complexity

Table 1. Example comparison of the state of the art.

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3. MATERIALS AND METHODS

The method was organized into four stages: system characterization, data consolidation, computational model development and validation. The first stage defines boundaries, assets, measurement points and variables of interest. The second addresses temporal consistency, units, missing values and traceability. The third establishes the hydraulic model and analytics layer. The fourth compares model outputs with independent observations using metrics defined in advance. To ensure consistency, each variable should be associated with its unit, source, update frequency and validation rule. This description avoids ambiguity and makes the study auditable by external readers.

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Each measurement point should be described by quantity, range, resolution, sampling frequency and functional location. When this information cannot be published because of industrial restrictions, the article should provide enough detail for readers to understand data quality and the limitations imposed on the study. In industrial applications, practical relevance should be accompanied by measurable performance criteria. Operational, energy or reliability gains need to be reported with values, intervals and operating conditions.

Preprocessing should not be treated as a neutral step. Filtering, smoothing, normalization and exclusion change data distributions and can influence model performance. Treatment decisions should therefore be justified and, when possible, evaluated through sensitivity analysis. The methodological structure should also record software versions, parameters and preprocessing decisions. These elements are essential for reproducibility and for an adequate assessment of the technological contribution.

In data-driven models, separation among training, validation and test sets should prevent temporal or operational leakage. Time series require particular care because random splits may place highly similar samples in different sets and produce optimistic performance estimates. To ensure consistency, each variable should be associated with its unit, source, update frequency and validation rule. This description avoids ambiguity and makes the study auditable by external readers.

3.1. System architecture

Figure 1 presents the architecture used as a visual reference. Each block represents a function and the arrows indicate information flow, not necessarily direct communication between devices.

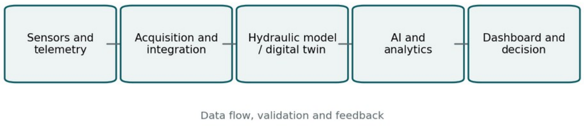

Figure 1. Integrated acquisition, modeling and decision architecture.

The instrumentation layer includes pressure, flow, level and equipment-state sensors. The integration layer normalizes timestamps and units. The physical model represents hydraulic relationships. Analytics computes indicators, and the decision layer presents alarms, scenarios and recommendations with traceability.

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When simulation is used, the manuscript should report initial and boundary conditions, integration step, convergence criteria and adjusted parameters. Validation should compare quantities directly related to the modeled phenomenon rather than only aggregated indicators that may hide local errors. The narrative should clearly separate observation, hypothesis and interpretation. When comparisons with previous work are made, the text must state the basis of comparison and avoid superiority claims without equivalent evidence.

3.2. Data, sampling and preprocessing

The illustrative dataset considers time series synchronized at five-minute intervals. Before analysis, records are checked for range, temporal consistency, duplicates and missing values. Short gaps may be interpolated when technically justified; long gaps should be flagged and excluded from analyses that depend on continuity. To ensure consistency, each variable should be associated with its unit, source, update frequency and validation rule. This description avoids ambiguity and makes the study auditable by external readers.

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3.3. Mathematical model and metrics

The estimated variable is expressed generically by Eq. (1), where x(t) is the input vector, θ is the parameter set and ε(t) is the error term.

$$\hat{y}(t) = f(x(t), \theta) + \varepsilon(t) \qquad (1)$$

For evaluation, metrics should be consistent with the problem. Regression studies may use MAE, RMSE and R^2; classification studies may use precision, recall, F1 and AUC. The manuscript should define the metric and justify its selection.

$$RMSE = \sqrt{[(1/n)\, \Sigma_i\, (y_i - \hat{y}_i)^2]} \qquad (2)$$

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Calibration should use a clearly defined subset of the data, while validation should rely on independent observations. When the system undergoes configuration changes, calibration and validation periods should be separated by operating condition. Results should include not only mean error but also dispersion, extreme values and any systematic trends. To ensure consistency, each variable should be associated with its unit, source, update frequency and validation rule. This description avoids ambiguity and makes the study auditable by external readers.

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3.5. Reproducibility

To make the study reproducible, the article should report software versions, parameters, random seeds when applicable, filtering criteria, dataset splits and the availability of code or data. Sharing restrictions should be justified by confidentiality, security, contractual or ethical constraints. To ensure consistency, each variable should be associated with its unit, source, update frequency and validation rule. This description avoids ambiguity and makes the study auditable by external readers.

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4. RESULTS

The results in this section are deliberately illustrative and demonstrate the expected reporting standard. The baseline condition was assigned a normalized error of 1.00. Three subsequent configurations produced errors of 0.74, 0.59 and 0.46. Relative reduction should be calculated against a defined baseline and accompanied by identification of the dataset used. To ensure consistency, each variable should be associated with its unit, source, update frequency and validation rule. This description avoids ambiguity and makes the study auditable by external readers.

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In addition to average performance, conditions under which the model fails should be examined. Errors concentrated in particular regimes may indicate physical behavior changes, insufficient data or the need for segmentation. This analysis is often more useful for engineering than a single global metric. The methodological structure should also record software versions, parameters and preprocessing decisions. These elements are essential for reproducibility and for an adequate assessment of the technological contribution.

When multiple configurations are compared, the baseline must be clearly defined. Percentage improvement is informative only when readers know which data, period and criteria were held constant. Simultaneous changes in architecture, sample and metric make it difficult to attribute improvement to a specific cause. To ensure consistency, each variable should be associated with its unit, source, update frequency and validation rule. This description avoids ambiguity and makes the study auditable by external readers.

4.1. Comparative performance

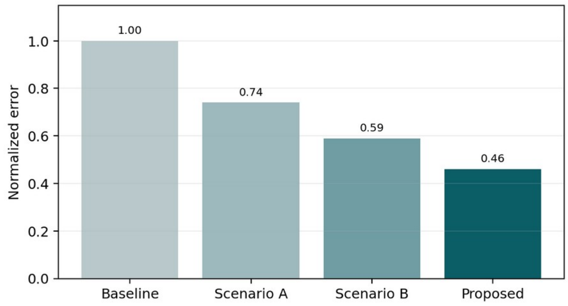

Figure 2. Example comparison of normalized error across scenarios.

Figure 2 shows a decreasing error trend, but interpretation must consider sample variability. In an actual article, the graph should be accompanied by confidence intervals, error bars or appropriate tests whenever statistical inference is part of the conclusion.

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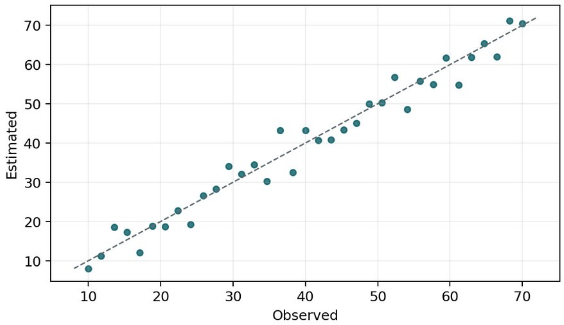
4.2. Validation of observed and estimated values

Figure 3 presents an illustrative relationship between observed and estimated values. Proximity to the 1:1 line suggests overall consistency, but distant points should be examined to identify specific operating conditions, sensor saturation or model limitations.



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4.3. Quantitative summary

Scenario	RMSE	MAE	R^2
Base	8,4	6,8	0,71
A	6,2	4,9	0,80
B	4,9	3,8	0,87
Proposed	3,9	3,0	0,92

Table 2. Example quantitative results. Illustrative values.

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5. DISCUSSION

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5.1. Interpretation of results

The error reduction in the proposed scenario may be interpreted as the combined effect of data consolidation and model-state updating. This interpretation should be supported by complementary analyses. If several modifications are introduced simultaneously, an ablation study or incremental comparison is recommended to separate the contribution of each component. To ensure consistency, each variable should be associated with its unit, source, update frequency and validation rule. This description avoids ambiguity and makes the study auditable by external readers.

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5.2. Comparison with previous work

Comparison with previous work should be performed on equivalent bases. Differences in scale, topology, period, sensors, error definition and data availability may prevent direct comparison. When quantitative comparison is not valid, the article should provide a qualitative comparison of architecture, requirements and limitations rather than turning different contexts into a performance ranking. To ensure consistency, each variable should be associated with its unit, source, update frequency and validation rule. This description avoids ambiguity and makes the study auditable by external readers.

The illustrative results indicate that integrating a physical model and data analytics can reduce estimation error compared with approaches based solely on fixed rules. Interpretation, however, must be conditioned on registry quality, sensor coverage and instrument stability. A technically consistent article should not turn an observed improvement into a universal claim; it should state the conditions under which the conclusion is valid and identify factors that may affect performance. The narrative should clearly separate observation, hypothesis and interpretation. When comparisons with previous work are made, the text must state the basis of comparison and avoid superiority claims without equivalent evidence.

The discussion should connect quantitative results to the engineering mechanism that may explain them. Statistical correlation alone does not demonstrate causality. When causal interpretation cannot be supported, the text should use language consistent with association or observational evidence. In industrial applications, practical relevance should be accompanied by measurable performance criteria. Operational, energy or reliability gains need to be reported with values, intervals and operating conditions.

Transferability also needs to be analyzed. A solution validated in one network, machine or plant may depend on local instrumentation, operation and maintenance characteristics. The article should state which elements are generalizable and which require recalibration before use in another context. The methodological structure should also record software versions, parameters and preprocessing decisions. These elements are essential for reproducibility and for an adequate assessment of the technological contribution.

Adoption cost is part of the discussion when the contribution is intended to support technology decisions. Computational resources, additional sensing needs, integration with existing systems and update effort may alter the relationship between benefit and complexity. Even without a complete economic analysis, these constraints should be acknowledged. To ensure consistency, each variable should be associated with its unit, source, update frequency and validation rule. This description avoids ambiguity and makes the study auditable by external readers.

5.3. Engineering implications

From an applied-engineering perspective, the architecture can support inspection prioritization, operating-scenario assessment and identification of inconsistencies between registry information and observed behavior. Benefits depend on integration with operation and maintenance routines. An automated recommendation that is not linked to an accountable person, confirmation rule and decision record tends to produce limited institutional value. To ensure consistency, each variable should be associated with its unit, source, update frequency and validation rule. This description avoids ambiguity and makes the study auditable by external readers.

Despite the progress of supervisory control and data acquisition platforms, field information, technical registries, maintenance histories and hydraulic models are still frequently fragmented. Fragmentation reduces decision traceability and makes it difficult to convert operational data into actionable knowledge. The research problem of this model article is therefore how to structure an architecture capable of integrating sensor data, a hydraulic model and artificial intelligence algorithms in an auditable, reproducible and decision-oriented manner. The narrative should clearly separate observation, hypothesis and interpretation. When comparisons with previous work are made, the text must state the basis of comparison and avoid superiority claims without equivalent evidence.

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Lifecycle governance should define who updates registries, validates changes, approves new models and tracks versions. Without this definition, the digital representation tends to progressively diverge from the physical asset, reducing confidence in analyses and decisions supported by the system. To ensure consistency, each variable should be associated with its unit, source, update frequency and validation rule. This description avoids ambiguity and makes the study auditable by external readers.

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6. IMPLEMENTATION AND APPLICATION

A deployment can begin with a limited set of critical assets and be expanded progressively. The first cycle should prioritize registry reliability, measurement availability and identifier integration. Analytical complexity should be expanded only after the data flow is stable. This sequence reduces the risk of building sophisticated models on inconsistent foundations. To ensure consistency, each variable should be associated with its unit, source, update frequency and validation rule. This description avoids ambiguity and makes the study auditable by external readers.

The method was organized into four stages: system characterization, data consolidation, computational model development and validation. The first stage defines boundaries, assets, measurement points and variables of interest. The second addresses temporal consistency, units, missing values and traceability. The third establishes the hydraulic model and analytics layer. The fourth compares model outputs with independent observations using metrics defined in advance. The narrative should clearly separate observation, hypothesis and interpretation. When comparisons with previous work are made, the text must state the basis of comparison and avoid superiority claims without equivalent evidence.

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Continuous operation requires mechanisms to monitor the model itself. Distribution shifts, sensor drift and process changes can degrade performance without producing an obvious failure. Model-health indicators and revalidation rules should be part of the operational architecture. The methodological structure should also record software versions, parameters and preprocessing decisions. These elements are essential for reproducibility and for an adequate assessment of the technological contribution.

Integration with end users should consider evidence presentation, confidence level and auditability. Recommendations should be understandable to professionals responsible for decisions and should preserve records of the information used, the recommendation issued and the action actually taken. To ensure consistency, each variable should be associated with its unit, source, update frequency and validation rule. This description avoids ambiguity and makes the study auditable by external readers.

6.1. Computational architecture and security

The computational architecture should separate acquisition, storage, processing and visualization functions. Interfaces with control systems must comply with security policies, least privilege, authentication and event logging. In industrial contexts, an analytics application should not be assumed to write directly to controllers or critical systems without specific governance. To ensure consistency, each variable should be associated with its unit, source, update frequency and validation rule. This description avoids ambiguity and makes the study auditable by external readers.

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6.2. Scalability, maintenance and cost

Scalability does not depend solely on computational capacity. The number of assets, update frequency, protocol heterogeneity and the need to validate registry information affect expansion cost. These factors should be discussed whenever the proposal is intended to extend beyond the initial case study. To ensure consistency, each variable should be associated with its unit, source, update frequency and validation rule. This description avoids ambiguity and makes the study auditable by external readers.

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7. LIMITATIONS

The main limitations of the example are associated with the illustrative nature of the data and the absence of a real operational deployment. Actual studies should report measurement uncertainty, seasonal changes, communication failures, topology changes, maintenance interventions and information-security constraints. The cost of model updates and the organizational effort required to maintain a reliable technical database should also be discussed. To ensure consistency, each variable should be associated with its unit, source, update frequency and validation rule. This description avoids ambiguity and makes the study auditable by external readers.

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8. CONCLUSIONS

This model article demonstrates a way to present applied engineering research with an explicit chain linking problem, method, results, discussion and conclusion. The proposed architecture organizes information into layers and shows that the value of a digital twin depends on data quality, model validation and integration with decision processes. In an actual submission, the conclusion should directly answer the objectives and highlight verified results, limitations and technically testable next steps. To ensure consistency, each variable should be associated with its unit, source, update frequency and validation rule. This description avoids ambiguity and makes the study auditable by external readers.

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8.1. Future work

As next steps, a real application could include validation under different operating regimes, integration with predictive maintenance and economic assessment of the model-update cycle. Future work should derive from identified limitations rather than generic lists unrelated to the results. To ensure consistency, each variable should be associated with its unit, source, update frequency and validation rule. This description avoids ambiguity and makes the study auditable by external readers.

DECLARATIONS

Author contributions (CRediT)

Conceptualization: Author 1; Methodology: Authors 1 and 2; Software: Author 2; Validation: Authors 2 and 3; Formal analysis: Authors 1 and 3; Investigation: all authors; Visualization: Author 2; Supervision: Author 3; Writing - original draft: Author 1; Writing - review and editing: all authors. All authors read and approved the final manuscript.

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Conflicts of interest

The authors declare no financial, personal or institutional conflicts of interest that could influence the conduct, interpretation or publication of this work. Any potential conflict should be objectively described.

Data and code availability

Data, scripts and code used in an actual article should be deposited in an appropriate repository, preferably with a persistent identifier. When sharing is not possible, access restrictions should be justified on legal, ethical, contractual, security or confidentiality grounds.

Use of generative AI

Objectively disclose any substantive use of generative AI in writing, programming, analysis, synthesis or image production. Tools cannot be listed as authors, and full responsibility for the content remains with the authors.

Ethics approval

When applicable, identify the committee, institution, protocol number, approval date and consent conditions. Research involving humans, animals or sensitive data should demonstrate ethical compliance; when approval is not applicable, explicitly state the justification.

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Supplementary material

Identify additional files, datasets, videos, models, code, drawings or technical documentation provided with the article. Supplementary materials should be cited in the text and have a clear identifier, version and direct relationship to the reported results.

REFERENCES

[1] A. A. Author and E. B. Author, “Article title 1: example journal reference,” Journal Title, vol. 9, no. 2, pp. 11-26, 2023. https://doi.org/10.xxxx/setari.ref01

[2] B. C. Author, Technical Book Title 2, 2nd ed. City: Publisher, 2024.

[3] C. D. Author et al., “Conference paper title 3,” in Proceedings of the International Engineering Conference, City, Country, 2025, pp. 3-9. https://doi.org/10.xxxx/conf03

[4] Technical Organization, “Technical standard or report 4,” Document No. 1004, 2026.

[5] E. E. Author, “Dataset 5,” Institutional Repository, version 1.1, 2022. https://doi.org/10.xxxx/data05

[6] F. F. Author, “Software or code 6,” version 1.0, Software Repository, 2023. https://doi.org/10.xxxx/code06

[7] G. A. Author and K. B. Author, “Article title 7: example journal reference,” Journal Title, vol. 15, no. 4, pp. 17-32, 2024. https://doi.org/10.xxxx/setari.ref07

[8] H. C. Author, Technical Book Title 8, 2nd ed. City: Publisher, 2025.

[9] I. D. Author et al., “Conference paper title 9,” in Proceedings of the International Engineering Conference, City, Country, 2026, pp. 9-15. https://doi.org/10.xxxx/conf09

[10] Technical Organization, “Technical standard or report 10,” Document No. 1010, 2022.

[11] K. E. Author, “Dataset 11,” Institutional Repository, version 1.3, 2023. https://doi.org/10.xxxx/data11

[12] L. F. Author, “Software or code 12,” version 1.0, Software Repository, 2024. https://doi.org/10.xxxx/code12

[13] M. A. Author and Q. B. Author, “Article title 13: example journal reference,” Journal Title, vol. 12, no. 2, pp. 23-38, 2025. https://doi.org/10.xxxx/setari.ref13

[14] N. C. Author, Technical Book Title 14, 2nd ed. City: Publisher, 2026.

[15] O. D. Author et al., “Conference paper title 15,” in Proceedings of the International Engineering Conference, City, Country, 2022, pp. 15-21. https://doi.org/10.xxxx/conf15

[16] Technical Organization, “Technical standard or report 16,” Document No. 1016, 2023.

[17] Q. E. Author, “Dataset 17,” Institutional Repository, version 1.1, 2024. https://doi.org/10.xxxx/data17

[18] R. F. Author, “Software or code 18,” version 1.0, Software Repository, 2025. https://doi.org/10.xxxx/code18

[19] S. A. Author and C. B. Author, “Article title 19: example journal reference,” Journal Title, vol. 9, no. 4, pp. 29-44, 2026. https://doi.org/10.xxxx/setari.ref19

[20] T. C. Author, Technical Book Title 20, 2nd ed. City: Publisher, 2022.

[21] A. D. Author et al., “Conference paper title 21,” in Proceedings of the International Engineering Conference, City, Country, 2023, pp. 21-27. https://doi.org/10.xxxx/conf21

[22] Technical Organization, “Technical standard or report 22,” Document No. 1022, 2024.

[23] C. E. Author, “Dataset 23,” Institutional Repository, version 1.3, 2025. https://doi.org/10.xxxx/data23

[24] D. F. Author, “Software or code 24,” version 1.0, Software Repository, 2026. https://doi.org/10.xxxx/code24

[25] E. A. Author and I. B. Author, “Article title 25: example journal reference,” Journal Title, vol. 15, no. 2, pp. 35-50, 2022. https://doi.org/10.xxxx/setari.ref25

[26] F. C. Author, Technical Book Title 26, 2nd ed. City: Publisher, 2023.

[27] G. D. Author et al., “Conference paper title 27,” in Proceedings of the International Engineering Conference, City, Country, 2024, pp. 27-33. https://doi.org/10.xxxx/conf27

[28] Technical Organization, “Technical standard or report 28,” Document No. 1028, 2025.

[29] I. E. Author, “Dataset 29,” Institutional Repository, version 1.1, 2026. https://doi.org/10.xxxx/data29

[30] J. F. Author, “Software or code 30,” version 1.0, Software Repository, 2022. https://doi.org/10.xxxx/code30

[31] K. A. Author and O. B. Author, “Article title 31: example journal reference,” Journal Title, vol. 12, no. 4, pp. 41-56, 2023. https://doi.org/10.xxxx/setari.ref31

[32] L. C. Author, Technical Book Title 32, 2nd ed. City: Publisher, 2024.

[33] M. D. Author et al., “Conference paper title 33,” in Proceedings of the International Engineering Conference, City, Country, 2025, pp. 33-39. https://doi.org/10.xxxx/conf33

[34] Technical Organization, “Technical standard or report 34,” Document No. 1034, 2026.

[35] O. E. Author, “Dataset 35,” Institutional Repository, version 1.3, 2022. https://doi.org/10.xxxx/data35

[36] P. F. Author, “Software or code 36,” version 1.0, Software Repository, 2023. https://doi.org/10.xxxx/code36

[37] Q. A. Author and A. B. Author, “Article title 37: example journal reference,” Journal Title, vol. 9, no. 2, pp. 47-62, 2024. https://doi.org/10.xxxx/setari.ref37

[38] R. C. Author, Technical Book Title 38, 2nd ed. City: Publisher, 2025.

[39] S. D. Author et al., “Conference paper title 39,” in Proceedings of the International Engineering Conference, City, Country, 2026, pp. 39-45. https://doi.org/10.xxxx/conf39

[40] Technical Organization, “Technical standard or report 40,” Document No. 1040, 2022.

[41] A. E. Author, “Dataset 41,” Institutional Repository, version 1.1, 2023. https://doi.org/10.xxxx/data41

[42] B. F. Author, “Software or code 42,” version 1.0, Software Repository, 2024. https://doi.org/10.xxxx/code42

REFERENCES - continued

[43] C. A. Author and G. B. Author, “Article title 43: example journal reference,” Journal Title, vol. 15, no. 4, pp. 53-68, 2025. https://doi.org/10.xxxx/setari.ref43

[44] D. C. Author, Technical Book Title 44, 2nd ed. City: Publisher, 2026.

[45] E. D. Author et al., “Conference paper title 45,” in Proceedings of the International Engineering Conference, City, Country, 2022, pp. 45-51. https://doi.org/10.xxxx/conf45

[46] Technical Organization, “Technical standard or report 46,” Document No. 1046, 2023.

[47] G. E. Author, “Dataset 47,” Institutional Repository, version 1.3, 2024. https://doi.org/10.xxxx/data47

[48] H. F. Author, “Software or code 48,” version 1.0, Software Repository, 2025. https://doi.org/10.xxxx/code48

[49] I. A. Author and M. B. Author, “Article title 49: example journal reference,” Journal Title, vol. 12, no. 2, pp. 59-74, 2026. https://doi.org/10.xxxx/setari.ref49

[50] J. C. Author, Technical Book Title 50, 2nd ed. City: Publisher, 2022.

[51] K. D. Author et al., “Conference paper title 51,” in Proceedings of the International Engineering Conference, City, Country, 2023, pp. 51-57. https://doi.org/10.xxxx/conf51

[52] Technical Organization, “Technical standard or report 52,” Document No. 1052, 2024.

[53] M. E. Author, “Dataset 53,” Institutional Repository, version 1.1, 2025. https://doi.org/10.xxxx/data53

[54] N. F. Author, “Software or code 54,” version 1.0, Software Repository, 2026. https://doi.org/10.xxxx/code54

[55] O. A. Author and S. B. Author, “Article title 55: example journal reference,” Journal Title, vol. 9, no. 4, pp. 65-80, 2022. https://doi.org/10.xxxx/setari.ref55

[56] P. C. Author, Technical Book Title 56, 2nd ed. City: Publisher, 2023.

[57] Q. D. Author et al., “Conference paper title 57,” in Proceedings of the International Engineering Conference, City, Country, 2024, pp. 57-63. https://doi.org/10.xxxx/conf57

[58] Technical Organization, “Technical standard or report 58,” Document No. 1058, 2025.

[59] S. E. Author, “Dataset 59,” Institutional Repository, version 1.3, 2026. https://doi.org/10.xxxx/data59

[60] T. F. Author, “Software or code 60,” version 1.0, Software Repository, 2022. https://doi.org/10.xxxx/code60

[61] A. A. Author and E. B. Author, “Article title 61: example journal reference,” Journal Title, vol. 15, no. 2, pp. 71-86, 2023. https://doi.org/10.xxxx/setari.ref61

[62] B. C. Author, Technical Book Title 62, 2nd ed. City: Publisher, 2024.

[63] C. D. Author et al., “Conference paper title 63,” in Proceedings of the International Engineering Conference, City, Country, 2025, pp. 63-69. https://doi.org/10.xxxx/conf63

[64] Technical Organization, “Technical standard or report 64,” Document No. 1064, 2026.

[65] E. E. Author, “Dataset 65,” Institutional Repository, version 1.1, 2022. https://doi.org/10.xxxx/data65

[66] F. F. Author, “Software or code 66,” version 1.0, Software Repository, 2023. https://doi.org/10.xxxx/code66

[67] G. A. Author and K. B. Author, “Article title 67: example journal reference,” Journal Title, vol. 12, no. 4, pp. 77-92, 2024. https://doi.org/10.xxxx/setari.ref67

[68] H. C. Author, Technical Book Title 68, 2nd ed. City: Publisher, 2025.

[69] I. D. Author et al., “Conference paper title 69,” in Proceedings of the International Engineering Conference, City, Country, 2026, pp. 69-75. https://doi.org/10.xxxx/conf69

[70] Technical Organization, “Technical standard or report 70,” Document No. 1070, 2022.

[71] K. E. Author, “Dataset 71,” Institutional Repository, version 1.3, 2023. https://doi.org/10.xxxx/data71

[72] L. F. Author, “Software or code 72,” version 1.0, Software Repository, 2024. https://doi.org/10.xxxx/code72

[73] M. A. Author and Q. B. Author, “Article title 73: example journal reference,” Journal Title, vol. 9, no. 2, pp. 83-98, 2025. https://doi.org/10.xxxx/setari.ref73

[74] N. C. Author, Technical Book Title 74, 2nd ed. City: Publisher, 2026.

[75] O. D. Author et al., “Conference paper title 75,” in Proceedings of the International Engineering Conference, City, Country, 2022, pp. 75-81. https://doi.org/10.xxxx/conf75

[76] Technical Organization, “Technical standard or report 76,” Document No. 1076, 2023.

[77] Q. E. Author, “Dataset 77,” Institutional Repository, version 1.1, 2024. https://doi.org/10.xxxx/data77

[78] R. F. Author, “Software or code 78,” version 1.0, Software Repository, 2025. https://doi.org/10.xxxx/code78

[79] S. A. Author and C. B. Author, “Article title 79: example journal reference,” Journal Title, vol. 15, no. 4, pp. 89-104, 2026. https://doi.org/10.xxxx/setari.ref79

[80] T. C. Author, Technical Book Title 80, 2nd ed. City: Publisher, 2022.

[81] A. D. Author et al., “Conference paper title 81,” in Proceedings of the International Engineering Conference, City, Country, 2023, pp. 81-87. https://doi.org/10.xxxx/conf81

[82] Technical Organization, “Technical standard or report 82,” Document No. 1082, 2024.

[83] C. E. Author, “Dataset 83,” Institutional Repository, version 1.3, 2025. https://doi.org/10.xxxx/data83

[84] D. F. Author, “Software or code 84,” version 1.0, Software Repository, 2026. https://doi.org/10.xxxx/code84

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References in this model are formatting examples and must be fully replaced by real sources cited in the manuscript. DOI should be provided as a full URL whenever available.

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